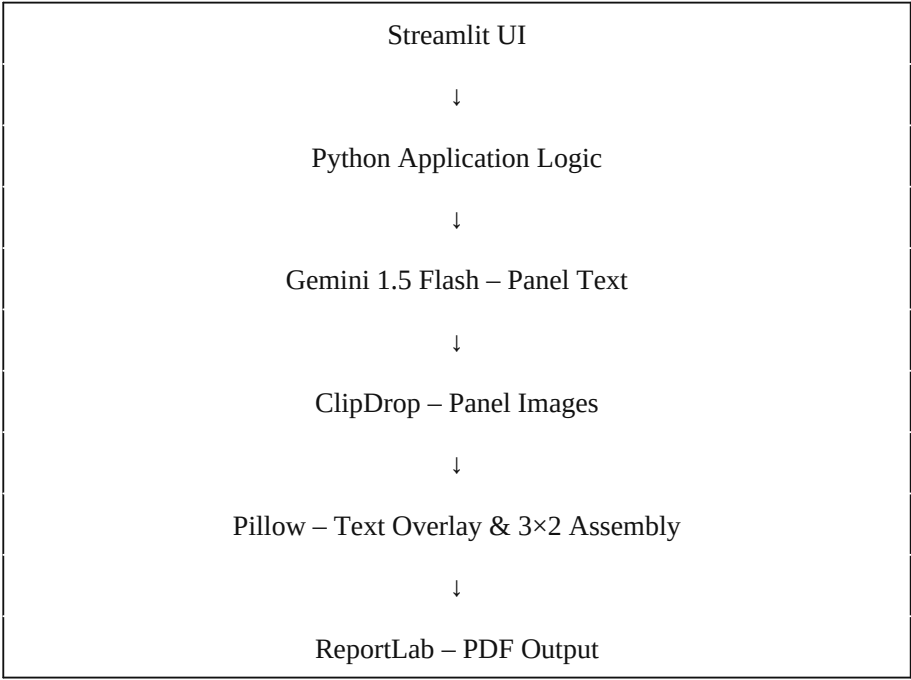


Solution Architecture

Date	03 October 2026
Team ID	COMICCRAFT-TEAM
Project Name	ComicCrafter AI – AI-Powered Comic Strip Generator
Maximum Marks	3 Marks

Architecture Overview



Module Responsibilities

Layer	Main Modules	Responsibility	Owner
UI / Integration	app.py	Collect prompt/style, run workflow, display output and PDF download.	Anwin Shellman J
AI Text	BACKEND/generate_panels.py	Gemini prompt, six-panel generation and response parsing.	Sivasankar C
AI Image	BACKEND/generate_image.py	Style mapping and ClipDrop image generation.	Sivasankaran M
Image Processing	BACKEND/process_comic.py	Borders, text wrapping and 3x2 comic-strip composition.	Sivasankaran M
Application / CLI	BACKEND/main.py	Command-line workflow for generation and assembly.	Anwin Shellman J
Output / QA	app.py, requirements.txt, README.md	PDF creation, download flow, setup and testing evidence.	Nishanth S

Data Flow

User enters a scenario and style → Streamlit invokes Gemini → six panel records are parsed → each description is sent to ClipDrop → six images are returned → Pillow adds dialogue and combines the panels → Streamlit previews the result → ReportLab creates the PDF.